COLLEGE CODE : 9628

COLLEGE NAME : University College of Engineering, Nagercoil

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IBM Cognos Analytics - Case Study Scenarios

Case Study Report : 10

ThreadCraft Textiles - Production & Sales Analytics

Industry : Garments & Textiles

Modules : Reporting, Exploration, Dashboard, Story Making

Production Efficiency Report

Question:1

Create a Production Efficiency Report showing Units Planned vs Units Produced vs Units Rejected by Factory Location and Garment Type. Add a calculated Efficiency % = (Produced / Planned) * 100. Apply Conditional Styling: Efficiency < 80% in Red, 80–90% in Amber, > 90% in Green. Add prompts for Factory and Season. Use Times New Roman font, size 11.

Procedure

Step 1: Create List Report

- Open IBM Cognos Analytics
- Create a **new List Report**
- Drag **Factory Location** and **Garment Type** from the source into the list

Step 2: Add Measures

- Insert measures:
  - Units Planned
  - Units Produced
  - Units Rejected

Step 3: Create Efficiency Calculation

- Drag a **Query Calculation** into the list
- Name it **Efficiency %**

Step 4: Enter Expression

- Expression:

  sql

  ([Units Produced] / [Units Planned]) * 100

- Format as **Percentage**

Step 5: Apply Conditional Styles

- Select the **Efficiency %** column

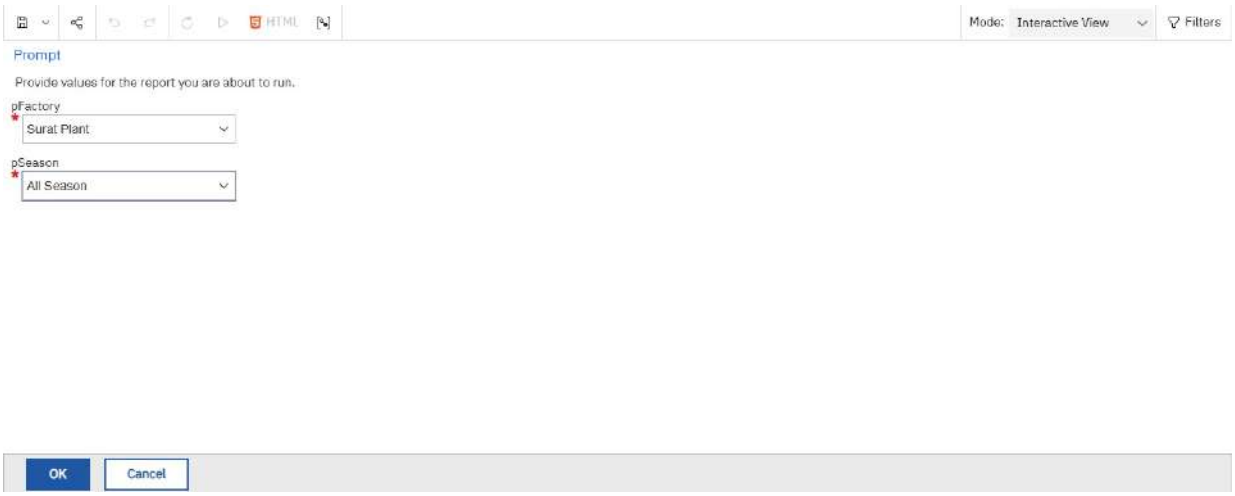
- Go to **Properties > Conditional Styles**
- Create a **New Advanced Conditional Style** with three rules:
    - If [Efficiency %] < 80 → Background = **Red**
    - If [Efficiency %] between 80 and 90 → Background = **Amber**
    - If [Efficiency %] > 90 → Background = **Green**

Step 6: Format List

- Select the entire list report
- Change **Font property** to *Times New Roman*, size **11**

Step 7: Add Filters

- Drag two **Value Prompts** to the header area:
    - **Factory** filter
    - **Season** filter
- Configure prompts for dynamic filtering



HTML Mode: Interactive View Filters

Garment Type	Factory Location	Units Planned	Units Produced	Units Rejected
Saree	Surat Plant	14,137	13,020	506
Kurti	Surat Plant	3,519	2,992	53
Jacket	Surat Plant	5,362	4,537	362
T-Shirt	Surat Plant	6,252	4,796	409
Kids Frock	Surat Plant	7,674	6,647	403
Jeans	Surat Plant	8,290	8,122	735
Trousers	Surat Plant	10,240	8,443	303
Formal Shirt	Surat Plant	7,607	6,015	224

Calculated Efficiency %
86.51

Export Order Fulfillment Report

Question:2

Build an Export Order Fulfillment Report showing Buyer Name, Order Quantity, Delivered Quantity, Fulfillment Rate % (calculated), and Delivery Status (On-Time/Delayed). Apply Conditional Styling: Delayed orders in Red bold. Add Header with ThreadCraft branding, Footer with 'Export Division Confidential'. Sort by Fulfillment Rate ascending to surface worst performers.

Procedure

Step 1: Create List Report

- Open IBM Cognos Analytics
- Create a **List Report**
- Add columns:
  - Buyer Name
  - Order Quantity
  - Delivered Quantity

Step 2: Fulfillment Rate Calculation

- Add a **Query Calculation** named *Fulfillment Rate %*:

  sql

  ```
  ([Delivered Quantity] / [Order Quantity]) * 100
  ```

- Format as **Percentage**

Step 3: Delivery Status Calculation

- Add another **Query Calculation** named *Delivery Status*:

  sql

  ```
  if ([Delivered Quantity] < [Order Quantity])
     then ('Delayed')
     else ('On-Time')
  ```

Step 4: Apply Conditional Styles

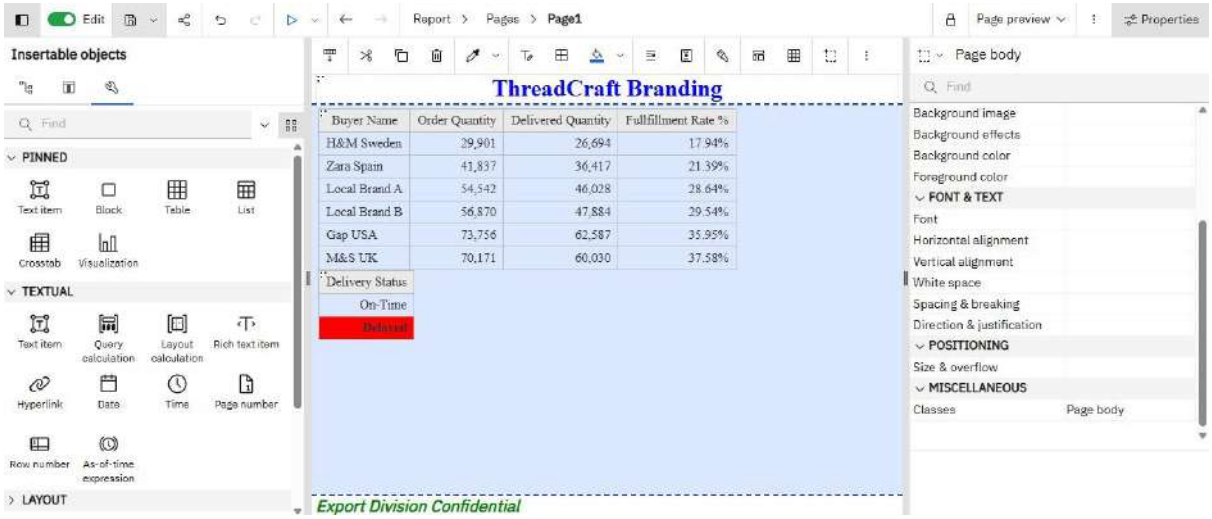
- Select the **Delivery Status** column
- Go to **Conditional Styles** in properties
- Add rule:
  - If [Delivery Status] = 'Delayed' → Font color = **Red**, Style = **Bold**

Step 5: Branding in Headers & Footers

- Open **Headers & Footers** menu
- Top Header → Type: *"ThreadCraft Branding"*
- Bottom Footer → Type: *"Export Division Confidential"*

Step 6: Sort Fulfillment Rate

- Select the **Fulfillment Rate %** column
- Click **Sort** → Choose **Ascending**
- Purpose: Show lowest fulfillment rates first



HTML Mode: Interactive View Filters

ThreadCraft Branding

Buyer Name	Order Quantity	Delivered Quantity	Fullfillment Rate %
H&M Sweden	29,901	26,694	17.94%
Zara Spain	41,837	36,417	21.39%
Local Brand A	54,542	46,028	28.64%
Local Brand B	56,870	47,884	29.54%
Gap USA	73,756	62,587	35.95%
M&S UK	70,171	60,030	37.58%

Delivery Status
On-Time
Delayed

Export Division Confidential

Interactive Sales Dashboard

Question:3

Using Report Visualize Prompt, allow Sales Managers to select chart type (Bar or Pie) to view Revenue by Season and Garment Type. Add prompts for Retail City and Season. Summarize the best-selling garment type and peak season revenue as text annotations. Use Conditional Styling to highlight products with revenue below ₹1,00,000 in Amber.

Procedure

Step 1: Create Variable

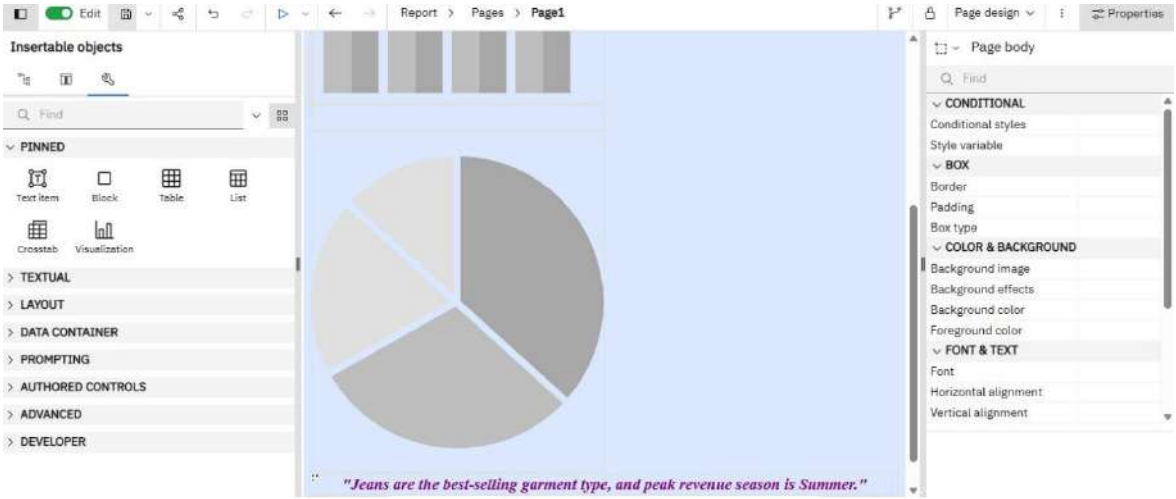
- Go to **Report > Variables**
- Create a **String Variable** named vChartToggle
- Expression: ParamValue('pChartType')

Step 2: Add Variable Values

- Add two values to vChartToggle:
  - **Bar**
  - **Pie**

Step 3: Add Charts

- On the report page, drag a **Bar Chart** and a **Pie Chart** into the same area
- Both charts use:
  - Category → Garment Type
  - Value → Revenue



Step 4: Configure Bar Chart

- Select the **Bar Chart**
- Go to **Render Variable property**
- Select vChartToggle
- Check only the **Bar** box

Step 5: Configure Pie Chart

- Select the **Pie Chart**
- Go to **Render Variable property**
- Select vChartToggle
- Check only the **Pie** box

Step 6: Add Prompts

- Drag three **Value Prompts** to the top:
  - **Prompt 1 (Chart Type):** Static choices → Bar, Pie → linked to parameter pChartType
  - **Prompt 2 (City):** Linked to data item → Multi-select = No → Auto-submit = Yes
  - **Prompt 3 (Season):** Linked to data item → Multi-select = No → Auto-submit = Yes

Step 7: Create Query Calculation

- Create a calculation named **ColorFlag**:

  sql

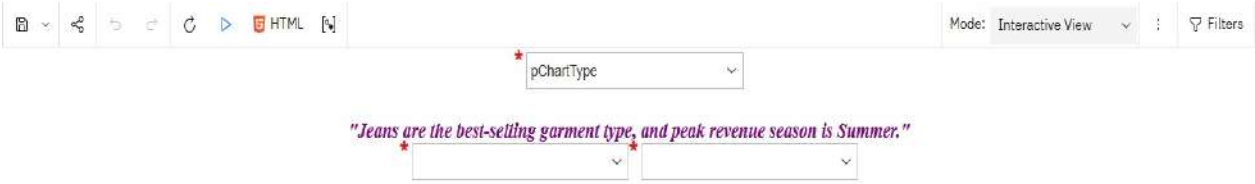
  ```
  if ([Revenue] < 100000) then ('Amber') else ('Normal')
  ```

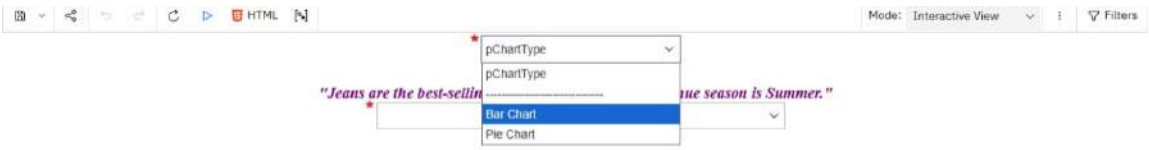
Step 8: Conditional Palette

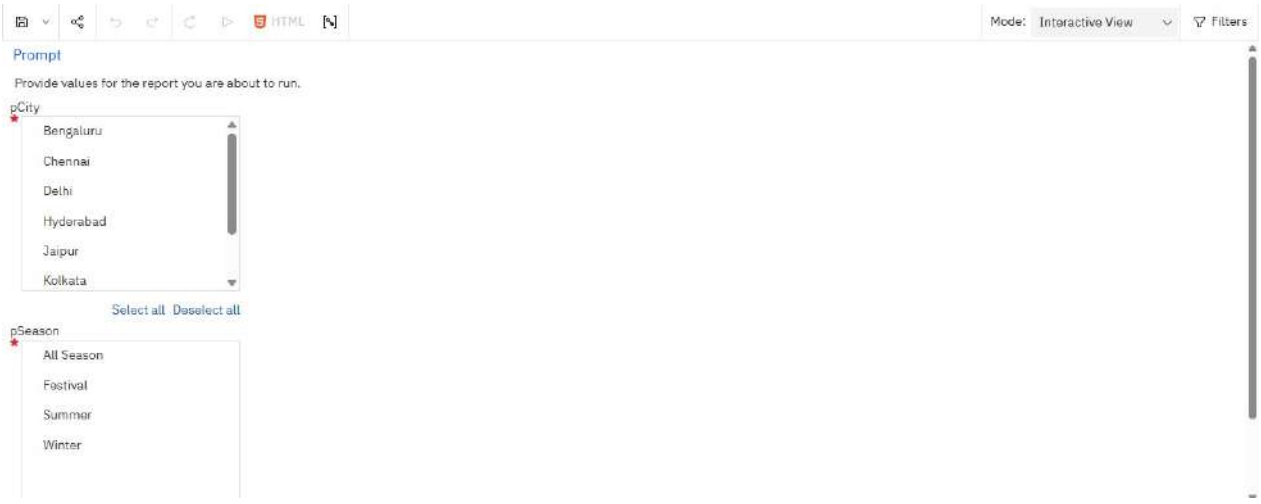
- Go to the **Bar Chart's Conditional Palette**
- Set color to **Amber** for the 'Amber' member

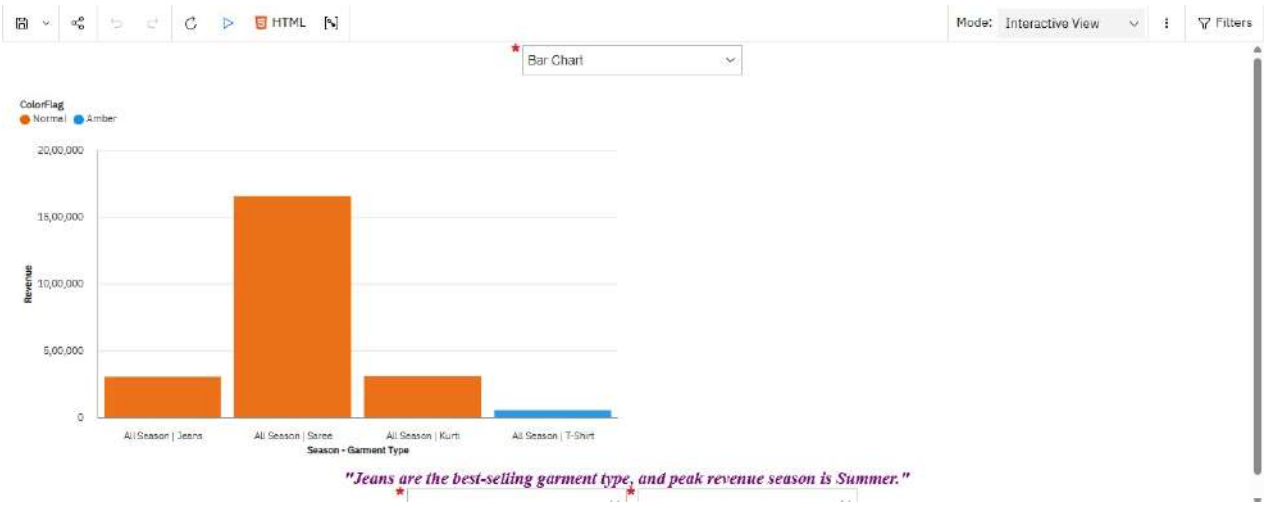
Step 9: Add Summary Text

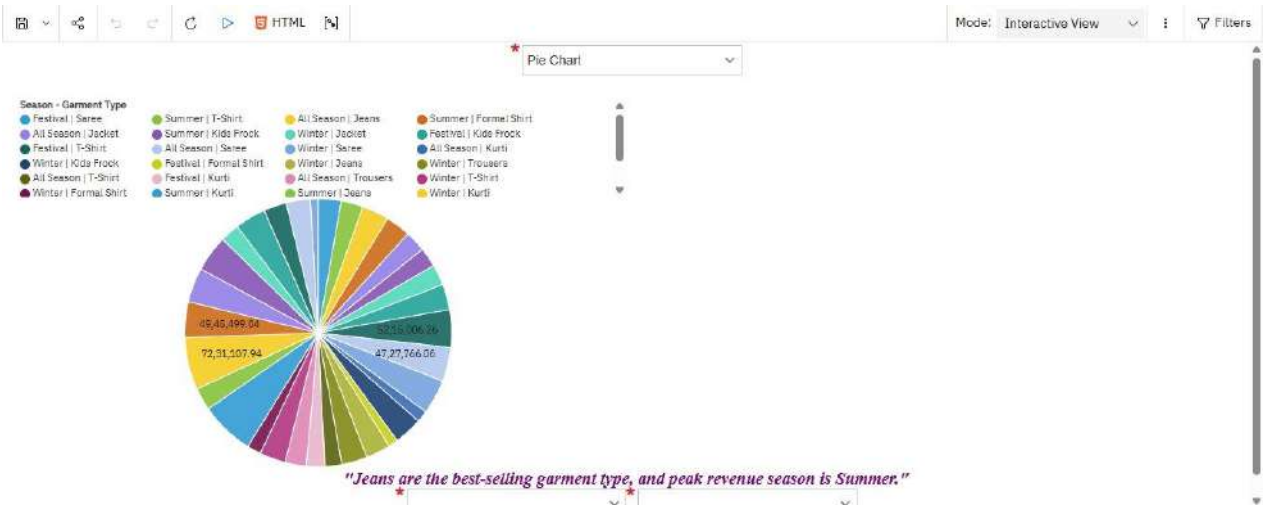
- Drag a **Text Item** to the bottom
- Type: *"Summary: Jeans are the best-selling garment type, and peak revenue season is Summer."*






HTML
Mode: Interactive View
Filters
Prompt
Provide values for the report you are about to run.
pCity
Bengaluru
Chennai
Delhi
Hydorabad
Jaipur
Kolkata
Select all Deselect all
pSeason
All Season
Festival
Summer
Winter


HTML
Mode: Interactive View
Filters
Bar Chart
ColorFlag
Normal
Amber
20,00,000
15,00,000
10,00,000
5,00,000
0
Revenue
All Season | Jeans
All Season | Saree
All Season | Kurti
All Season | T-Shirt
Season - Garment Type
"Jeans are the best-selling garment type, and peak revenue season is Summer."


HTML
Mode: Interactive View
Filters
Pie Chart
Season - Garment Type
Festival | Saree
Summer | T-Shirt
All Season | Jeans
Summer | Formal Shirt
All Season | Jacket
Summer | Kids Frock
Winter | Jacket
Festival | Kids Frock
Festival | T-Shirt
All Season | Saree
Winter | Saree
All Season | Kurti
Winter | Kids Frock
Festival | Formal Shirt
Winter | Jeans
Winter | Trousers
All Season | T-Shirt
Festival | Kurti
All Season | Trousers
Winter | T-Shirt
Winter | Formal Shirt
Summer | Kurti
Summer | Jeans
Winter | Kurti
45,45,499.04
72,31,107.94
47,27,766.06
"Jeans are the best-selling garment type, and peak revenue season is Summer."

Fabric Waste Exploration

Question:4

Use Cognos Exploration to explore the relationship between Fabric Wasted and Units Rejected. Investigate whether certain Fabric Types or Factories have significantly higher waste rates. Also explore how Season and Garment Type influence Retail Sales Volume. Document your key findings with annotated charts.

Procedure

Step 1: Data Upload

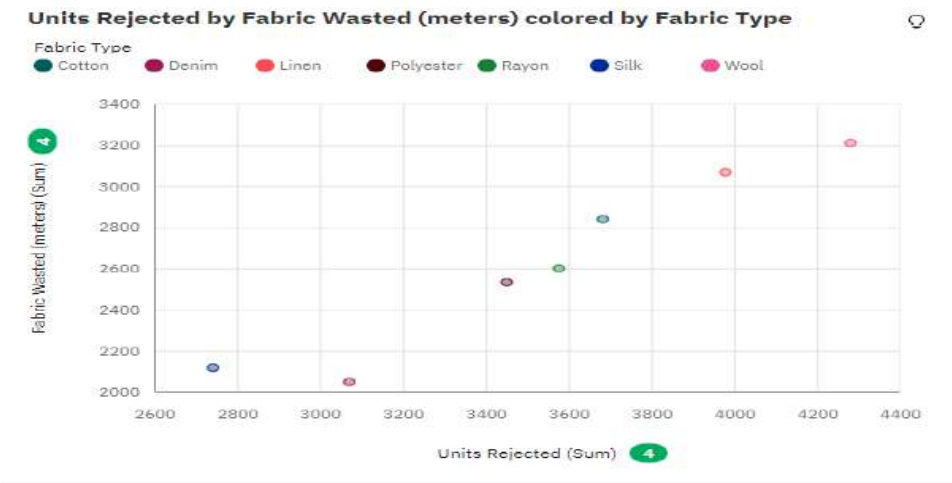
- Uploaded dataset into **IBM Cognos Analytics** using the **Upload Files** option
- Saved the file under **My Content**

Step 2: Creating Exploration

- Created a new **Exploration** using the uploaded dataset
- Opened dataset in the **Exploration interface** for analysis

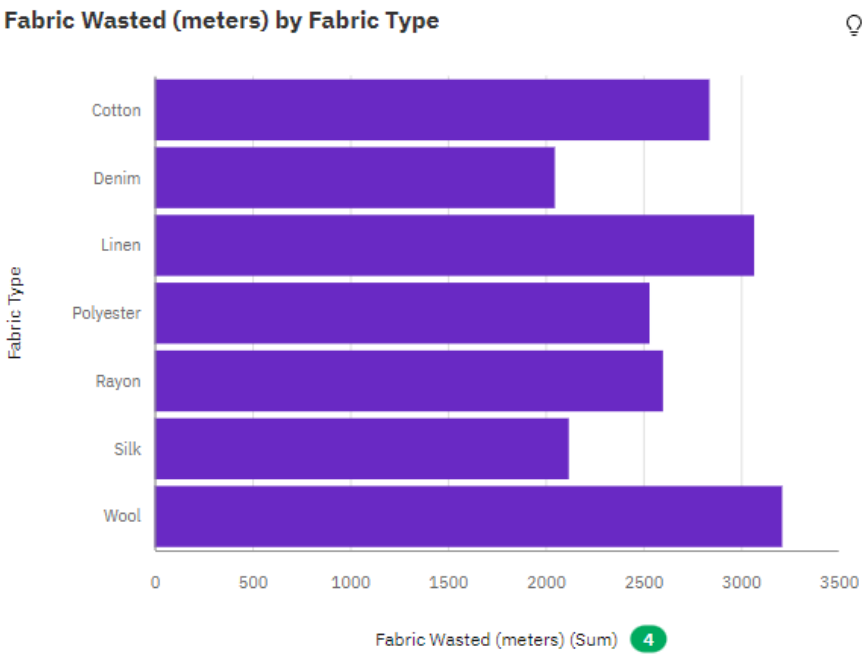
Step 3: Relationship Analysis (Scatter Plot)

- Added **Scatter Plot** visualization
- X-axis → Units Rejected
- Y-axis → Fabric Wasted
- Color → Fabric Type
- Details → Factory Location
- Purpose: Identify correlation between rejection and waste



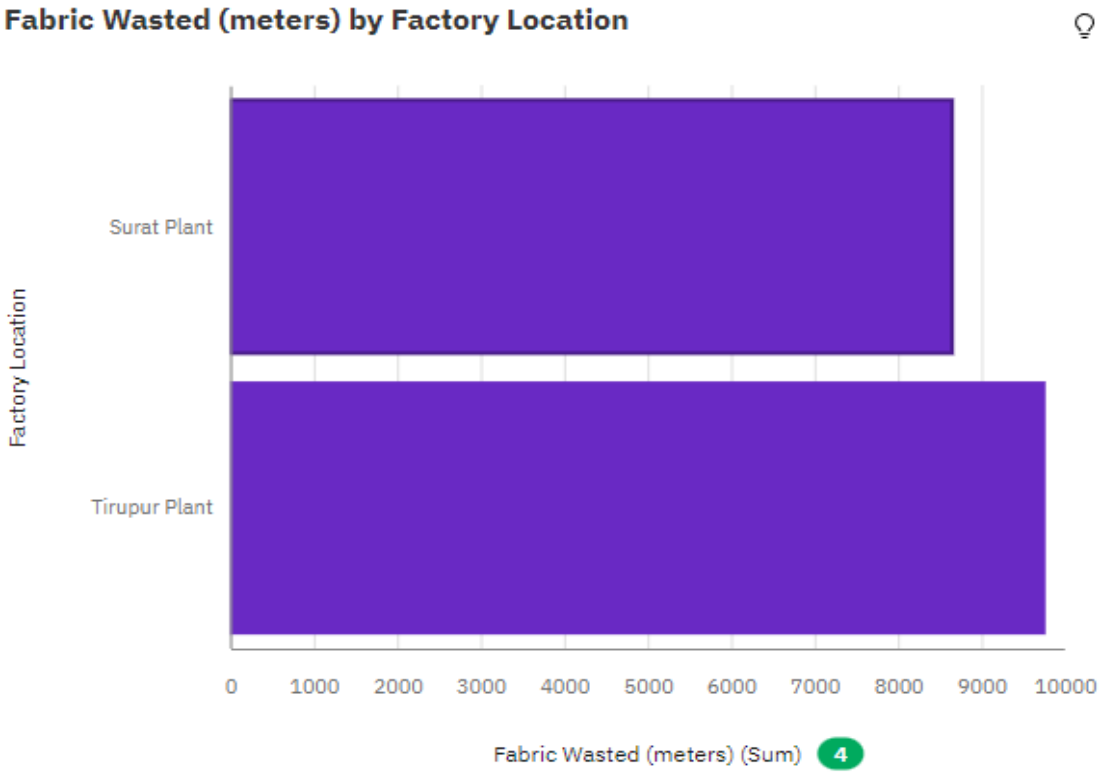
Step 4: Fabric Type Analysis (Bar Chart)

- Added **Bar Chart**
- X-axis → Fabric Type
- Y-axis → Fabric Wasted (SUM)
- Purpose: Compare waste levels across fabric types



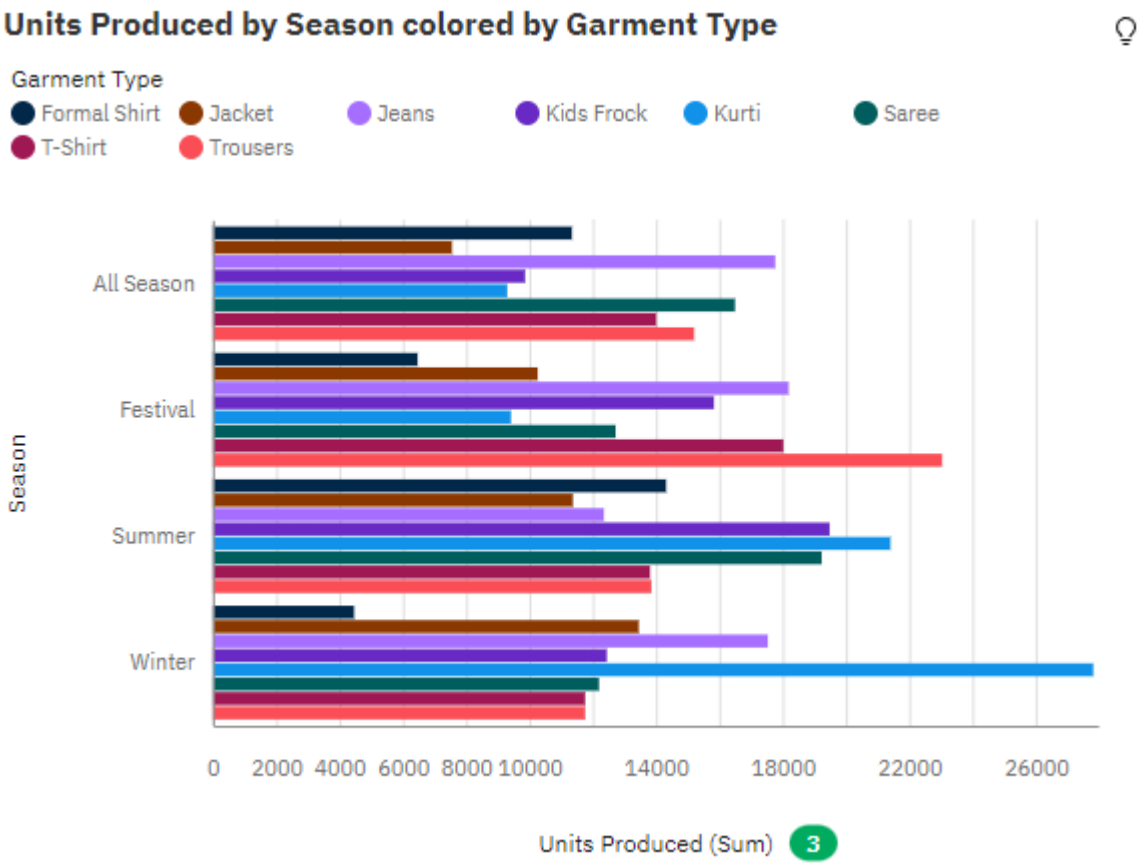
Step 5: Factory Analysis (Bar Chart)

- Added **Bar Chart**
- X-axis → Factory Location
- Y-axis → Fabric Wasted (SUM)
- Purpose: Identify factories with higher waste levels



Step 6: Seasonal Analysis (Bar Chart)

- Added **Bar Chart**
- X-axis → Season
- Y-axis → Units Produced
- Color → Garment Type
- Note: Retail Sales Volume not available, so Units Produced used as alternative
- Purpose: Show seasonal production trends



Step 7: Annotation

- Added **Insights and Observations** to each chart using Cognos annotation features
- Purpose: Provide context and highlight key findings

ThreadCraft Operations Dashboard

Question:5

Design a ThreadCraft Operations Dashboard with: (1) KPI tiles — Total Units Produced, Production Efficiency %, Fabric Waste %, Export Fulfillment Rate, (2) A Bar chart for Revenue by Garment Type, (3) A Line chart for Monthly Production Trend, (4) A Pie chart for Order Type Split (Domestic vs Export), (5) An Alert table for orders below 80% fulfillment. Add Factory and Season filters.

Procedure

Step 1: Load Dataset

- Open IBM Cognos Analytics
- Click **New → Dashboard**
- Load dataset: **Garment_ThreadCraft.xlsx**
- Ensure fields available: Units Produced, Revenue, Order Quantity, Delivery Quantity, Factory, Season, etc.

Step 2: Create Calculated Fields

Created required calculations:

- **Production Efficiency %**

Units Produced / Planned Units

- **Fabric Waste %**

Fabric Wasted / Fabric Used

- **Export Fulfillment Rate**

Export Delivered / Export Ordered

Formatted all values as percentages.

Step 3: Add KPI Tiles

Inserted KPI widgets for:

- Total Units Produced

- Production Efficiency %
- Fabric Waste %
- Export Fulfillment Rate

Applied formatting and labels.

Step 4: Create Bar Chart

- Added **Bar Chart**
- Category → Garment Type
- Value → Revenue (SUM)
- Displays revenue distribution across garment types

Step 5: Create Line Chart

- Added **Line Chart**
- X-axis → Month
- Y-axis → Units Produced
- Shows monthly production trend

Step 6: Create Pie Chart

- Added **Pie Chart**
- Category → Order Type (Domestic / Export)
- Value → Order Quantity (Count)
- Displays order type distribution

Step 7: Create Alert Table

- Added **Table widget**
- Columns: Order ID, Factory Location, Garment Type, Order Quantity, Delivery Quantity
- Created **Fulfillment %**:

Delivery Quantity / Order Quantity

- Applied filter: Fulfillment % < 0.8
- Displays only orders below 80% fulfillment → acts as alert table

Step 8: Add Filters

- Added **Factory Location filter**
    - Dragged Factory field → applied filter
- Added **Season filter**

- Dragged Season field → applied filter
- Connected filters to: KPI tiles, Bar chart, Line chart, Pie chart, Table
- Ensured dashboard updates dynamically

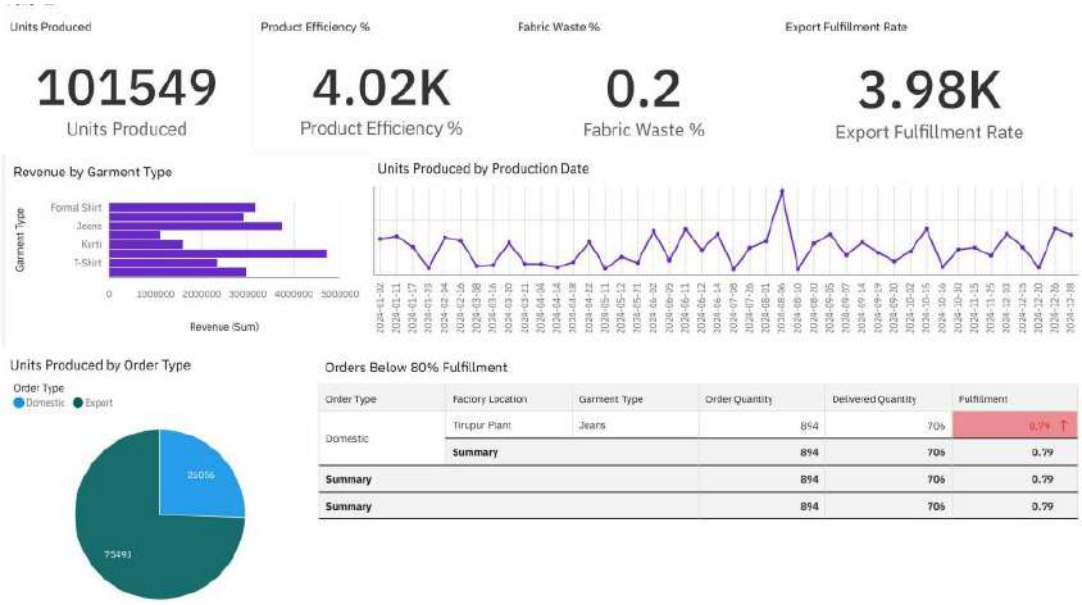
Step 9: Arrange Dashboard Layout

- **Top** → KPI tiles
- **Middle** → Bar chart and Line chart
- **Bottom** → Pie chart and Alert table
- Filters placed at top for easy access

Step 10: Final Formatting

Applied:

- Percentage formatting
- Titles and labels



HTML Code :

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>InsightHub | IBM Cognos Analytics Platform</title>
<style>
    * {
        margin: 0;
        padding: 0;
        box-sizing: border-box;
        font-family: 'Inter', -apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, sans-serif;
    }

    body {
        background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
        min-height: 100vh;
    }

    /* Login Page */
    .login-container {
        min-height: 100vh;
        display: flex;
        align-items: center;
        justify-content: center;
        padding: 1rem;
    }

    .login-card {
        background: white;
        border-radius: 2rem;
        padding: 2.5rem;
        width: 100%;
        max-width: 450px;
```

```
    box-shadow: 0 25px 50px -12px rgba(0, 0, 0, 0.25);
    animation: fadeIn 0.5s ease-out;
}

.login-header {
    text-align: center;
    margin-bottom: 2rem;
}

.login-icon {
    font-size: 3rem;
    margin-bottom: 0.5rem;
}

.login-title {
    font-size: 1.8rem;
    font-weight: 700;
    background: linear-gradient(135deg, #667eea, #764ba2);
    -webkit-background-clip: text;
    background-clip: text;
    color: transparent;
}

.login-subtitle {
    color: #6b7280;
    font-size: 0.875rem;
    margin-top: 0.25rem;
}
```

```
.input-group {
    margin-bottom: 1.25rem;
}

.input-group label {
    display: block;
    font-size: 0.8rem;
    font-weight: 600;
    color: #374151;
    margin-bottom: 0.5rem;
}

.input-field {
    width: 100%;
    padding: 0.75rem 1rem;
    border: 1px solid #e5e7eb;
    border-radius: 1rem;
    font-size: 1rem;
    transition: all 0.2s;
    outline: none;
}

.input-field:focus {
    border-color: #667eea;
    box-shadow: 0 0 0 3px rgba(102, 126, 234, 0.1);
}

.login-btn {
    width: 100%;
```

```
    padding: 0.875rem;
    background: linear-gradient(135deg, #667eea, #764ba2);
    color: white;
    border: none;
    border-radius: 1rem;
    font-size: 1rem;
    font-weight: 600;
    cursor: pointer;
    transition: transform 0.2s;
}

.login-btn:hover {
    transform: translateY(-2px);
}

.demo-cred {
    background: #f3f4f6;
    padding: 0.75rem;
    border-radius: 1rem;
    font-size: 0.75rem;
    text-align: center;
    margin-top: 1.5rem;
    color: #4b5563;
}

.error-msg {
    background: #fee2e2;
    color: #dc2626;
    padding: 0.75rem;
```

```
    border-radius: 1rem;
    font-size: 0.8rem;
    margin-top: 1rem;
    text-align: center;
    display: none;
}

/* Dashboard Layout */
.dashboard-container {
    min-height: 100vh;
    background: #f9fafb;
}

.navbar {
    background: white;
    padding: 1rem 2rem;
    display: flex;
    justify-content: space-between;
    align-items: center;
    border-bottom: 1px solid #e5e7eb;
    box-shadow: 0 1px 3px rgba(0,0,0,0.05);
}

.navbar-brand {
    display: flex;
    align-items: center;
    gap: 0.5rem;
}
```

```
.navbar-brand span {
    font-size: 1.5rem;
}

.navbar-brand h1 {
    font-size: 1.25rem;
    font-weight: 600;
    background: linear-gradient(135deg, #667eea, #764ba2);
    -webkit-background-clip: text;
    background-clip: text;
    color: transparent;
}

.user-info {
    display: flex;
    align-items: center;
    gap: 1rem;
}

.user-name {
    padding: 0.5rem 1rem;
    background: #f3f4f6;
    border-radius: 2rem;
    font-size: 0.875rem;
    font-weight: 500;
}

.logout-btn {
    padding: 0.5rem 1rem;
```

```
    background: white;
    border: 1px solid #e5e7eb;
    border-radius: 2rem;
    cursor: pointer;
    transition: all 0.2s;
}

.logout-btn:hover {
    background: #f3f4f6;
}

/* Main Content */
.main-content {
    padding: 2rem;
    max-width: 1400px;
    margin: 0 auto;
}

.welcome-banner {
    background: linear-gradient(135deg, #667eea, #764ba2);
    border-radius: 1.5rem;
    padding: 1.5rem 2rem;
    color: white;
    margin-bottom: 2rem;
}

.welcome-banner h2 {
    font-size: 1.5rem;
    margin-bottom: 0.5rem;
```

```
}

.welcome-banner p {
    opacity: 0.9;
}

/* Dataset Card */
.dataset-card {
    background: white;
    border-radius: 1.5rem;
    padding: 1.5rem;
    margin-bottom: 1.5rem;
    box-shadow: 0 1px 3px rgba(0,0,0,0.1);
    border: 1px solid #e5e7eb;
}

.dataset-header {
    display: flex;
    justify-content: space-between;
    align-items: center;
    margin-bottom: 1rem;
    padding-bottom: 1rem;
    border-bottom: 1px solid #e5e7eb;
}

.dataset-title {
    display: flex;
    align-items: center;
    gap: 0.75rem;
```

```
}

.dataset-icon {
    font-size: 1.5rem;
}

.dataset-title h3 {
    font-size: 1.25rem;
    color: #1f2937;
}

.dataset-badge {
    background: #dbeafe;
    color: #1e40af;
    padding: 0.25rem 0.75rem;
    border-radius: 2rem;
    font-size: 0.75rem;
    font-weight: 600;
}

.dataset-info {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
    gap: 1rem;
    margin-bottom: 1.5rem;
}

.info-item {
    background: #f9fafb;
```

```
    padding: 0.75rem;
    border-radius: 1rem;
}

.info-label {
    font-size: 0.7rem;
    text-transform: uppercase;
    color: #6b7280;
    font-weight: 600;
}

.info-value {
    font-size: 0.9rem;
    color: #1f2937;
    margin-top: 0.25rem;
    font-family: monospace;
}

/* Action Buttons */
.action-buttons {
    display: flex;
    gap: 1rem;
    flex-wrap: wrap;
}

.btn-primary {
    background: linear-gradient(135deg, #667eea, #764ba2);
    color: white;
    padding: 0.75rem 1.5rem;
```

```
    border: none;
    border-radius: 1rem;
    font-weight: 600;
    cursor: pointer;
    display: inline-flex;
    align-items: center;
    gap: 0.5rem;
    transition: transform 0.2s;
    text-decoration: none;
}

.btn-primary:hover {
    transform: translateY(-2px);
}

.btn-secondary {
    background: white;
    color: #4b5563;
    padding: 0.75rem 1.5rem;
    border: 1px solid #e5e7eb;
    border-radius: 1rem;
    font-weight: 600;
    cursor: pointer;
    display: inline-flex;
    align-items: center;
    gap: 0.5rem;
    transition: all 0.2s;
}
```

```
.btn-secondary:hover {
    background: #f9fafb;
    border-color: #d1d5db;
}

/* Help Section */
.help-card {
    background: #fefce8;
    border: 1px solid #fef08a;
    border-radius: 1rem;
    padding: 1rem 1.5rem;
    margin-top: 1.5rem;
}

.help-title {
    font-weight: 600;
    color: #854d0e;
    margin-bottom: 0.5rem;
}

.help-text {
    font-size: 0.875rem;
    color: #713f12;
    line-height: 1.5;
}

.help-text code {
    background: #fef3c7;
    padding: 0.125rem 0.375rem;
```

```
    border-radius: 0.375rem;
    font-family: monospace;
}

@keyframes fadeIn {
    from {
        opacity: 0;
        transform: translateY(20px);
    }
    to {
        opacity: 1;
        transform: translateY(0);
    }
}

.hidden {
    display: none !important;
}

@media (max-width: 768px) {
    .main-content {
        padding: 1rem;
    }
    .navbar {
        padding: 0.75rem 1rem;
    }
    .dataset-info {
        grid-template-columns: 1fr;
    }
```

```
        }
    </style>
</head>
<body>


<!-- LOGIN PAGE -->
<div id="loginPage" class="login-container">
    <div class="login-card">
        <div class="login-header">
            <div class="login-icon">📊</div>
            <div class="login-title">InsightHub</div>
            <div class="login-subtitle">Enterprise Analytics Platform</div>
        </div>
        <form id="loginForm">
            <div class="input-group">
                <label>Email Address</label>
                <input type="email" id="username" class="input-field" placeholder="analyst@company.com" autocomplete="off">
            </div>
            <div class="input-group">
                <label>Password</label>
                <input type="password" id="password" class="input-field" placeholder="••••••••">
            </div>
            <button type="submit" class="login-btn">Sign In →</button>
            <div id="loginError" class="error-msg"></div>
        </form>
        <div class="demo-cred">
            🔐 Demo Access: <strong>admin@insighthub.com</strong> / <strong>admin123</strong>
        </div>
    </div>
```

```
</div>


<!-- DASHBOARD PAGE -->
<div id="dashboardPage" class="dashboard-container hidden">
    <nav class="navbar">
        <div class="navbar-brand">
            <span>📈</span>
            <h1>InsightHub Analytics</h1>
        </div>
        <div class="user-info">
            <span class="user-name" id="displayUserName">User</span>
            <button id="logoutBtn" class="logout-btn">Sign Out</button>
        </div>
    </nav>


    <div class="main-content">
        <div class="welcome-banner">
            <h2>Welcome back, <span id="welcomeName">Analyst</span>! 👋</h2>
            <p>Your IBM Cognos Ques+ dataset is ready for analysis</p>
        </div>


        <!-- Ques+ Dataset Card -->
        <div class="dataset-card">
            <div class="dataset-header">
                <div class="dataset-title">
                    <span class="dataset-icon">📊</span>
                    <h3>Ques+ Dataset</h3>
                </div>
                <span class="dataset-badge">IBM Cognos Analytics</span>
```

```
</div>

<div class="dataset-info">
    <div class="info-item">
        <div class="info-label">Dataset Name</div>
        <div class="info-value">ques+5</div>
    </div>
    <div class="info-item">
        <div class="info-label">Platform</div>
        <div class="info-value">IBM Cognos Analytics</div>
    </div>
    <div class="info-item">
        <div class="info-label">Server</div>
        <div class="info-value">ap2.ca.analytics.ibm.com</div>
    </div>
    <div class="info-item">
        <div class="info-label">Dashboard ID</div>
        <div class="info-value">model0000019d686a732f_00000000</div>
    </div>
</div>

<div class="action-buttons">
    <button id="openCognosBtn" class="btn-primary">
        🚀 Open in IBM Cognos
    </button>
    <button id="copyUrlBtn" class="btn-secondary">
        📋 Copy Dashboard URL
    </button>
    <button id="refreshDatasetBtn" class="btn-secondary">
```

```
            🔄 Refresh Dataset Info
        </button>
    </div>
</div>

<!-- Help / Troubleshooting Card -->
<div class="help-card">
    <div class="help-title">ℹ About IBM Cognos Integration</div>
    <div class="help-text">
        The Ques+ dataset dashboard is hosted on IBM Cognos Analytics Cloud.
        Click the <strong>"Open in IBM Cognos"</strong> button above to view your
        full interactive dashboard in a new tab. <br><br>
        <strong>Why not embedded?</strong> IBM Cognos iframes require specific
        authentication cookies and CORS settings. Opening in a new tab ensures
        full functionality with all filters, visualizations, and export options.
    </div>
</div>

<!-- Quick Stats Preview -->
<div class="dataset-card" style="margin-top: 1rem; background: linear-gradient(135deg, #f0fdf4, #dcfce7);">
    <div class="dataset-header">
        <div class="dataset-title">
            <span class="dataset-icon">✅</span>
            <h3>Connection Status</h3>
        </div>
        <span class="dataset-badge" style="background: #dcfce7; color: #166534;">Active</span>
    </div>
    <div style="display: flex; gap: 2rem; flex-wrap: wrap; margin-top: 0.5rem;">
        <div>
```

```
                <div style="font-size: 0.7rem; color: #166534;">Status</div>
                <div style="font-weight: 600;">✓ Cognos Access Ready</div>
            </div>
            <div>
                <div style="font-size: 0.7rem; color: #166534;">Dataset</div>
                <div style="font-weight: 600;">ques+5 · Loaded</div>
            </div>
            <div>
                <div style="font-size: 0.7rem; color: #166534;">Action Required</div>
                <div style="font-weight: 600;">Click button above to view</div>
            </div>
        </div>
    </div>
  </div>
</div>

<script>
  // DOM Elements
  const loginPage = document.getElementById('loginPage');
  const dashboardPage = document.getElementById('dashboardPage');
  const loginForm = document.getElementById('loginForm');
  const loginError = document.getElementById('loginError');
  const logoutBtn = document.getElementById('logoutBtn');
  const displayUserName = document.getElementById('displayUserName');
  const welcomeName = document.getElementById('welcomeName');
  const openCognosBtn = document.getElementById('openCognosBtn');
  const copyUrlBtn = document.getElementById('copyUrlBtn');
  const refreshDatasetBtn = document.getElementById('refreshDatasetBtn');
```

```
// IBM Cognos Dashboard URL (your exact iframe source)

const COGNOS_URL =
"https://ap2.ca.analytics.ibm.com/bi/?perspective=dashboard&pathRef=.my_folders%2Fques%2B5&closeWindowOnLastVie
w=true&ui_appbar=false&ui_navbar=false&shareMode=embedded&action=view&mode=dashboard&subView=model000001
9d686a732f_00000000&nav_filter=true";



// Simple authentication (mock)

const validateCredentials = (email, password) => {

    if (!email || !password) return false;

    const validUsers = [

        'admin@insighthub.com',

        'analyst@insighthub.com',

        'demo@insighthub.com'

    ];

    return validUsers.includes(email.trim()) && password.trim().length >= 4;

};



const checkAuth = () => {

    const isLoggedIn = sessionStorage.getItem('insightAuth') === 'true';

    const savedUser = sessionStorage.getItem('insightUser') || '';

    if (isLoggedIn && savedUser) {

        showDashboard(savedUser);

    } else {

        showLogin();

    }

};



const showLogin = () => {

    loginPage.classList.remove('hidden');

    dashboardPage.classList.add('hidden');
```

```
    if (loginError) loginError.style.display = 'none';
};

const showDashboard = (username) => {
    const displayName = username.split('@')[0];
    if (displayUserName) displayUserName.textContent = `👤 ${displayName}`;
    if (welcomeName) welcomeName.textContent = displayName;

    loginPage.classList.add('hidden');
    dashboardPage.classList.remove('hidden');
};

// Open Cognos in new tab
const openCognosDashboard = () => {
    window.open(COGNOS_URL, '_blank', 'noopener,noreferrer');
    // Show feedback
    const btn = openCognosBtn;
    const originalText = btn.innerHTML;
    btn.innerHTML = "✅ Opened in new tab!";
    setTimeout(() => {
        btn.innerHTML = originalText;
    }, 2000);
};

// Copy URL to clipboard
const copyDashboardUrl = async () => {
    try {
        await navigator.clipboard.writeText(COGNOS_URL);
        const btn = copyUrlBtn;
```

```
        const originalText = btn.innerHTML;
        btn.innerHTML = "✅ Copied!";
        setTimeout(() => {
            btn.innerHTML = originalText;
        }, 2000);
    } catch (err) {
        alert('Could not copy. Select and copy the URL manually.');
    }
};

// Refresh dataset info (just visual feedback)
const refreshDatasetInfo = () => {
    const btn = refreshDatasetBtn;
    const originalText = btn.innerHTML;
    btn.innerHTML = "🔄 Refreshing...";
    setTimeout(() => {
        btn.innerHTML = "✅ Dataset verified!";
        setTimeout(() => {
            btn.innerHTML = originalText;
        }, 1500);
    }, 500);
};

// Login form handler
loginForm.addEventListener('submit', (e) => {
    e.preventDefault();
    const username = document.getElementById('username').value;
    const password = document.getElementById('password').value;
```

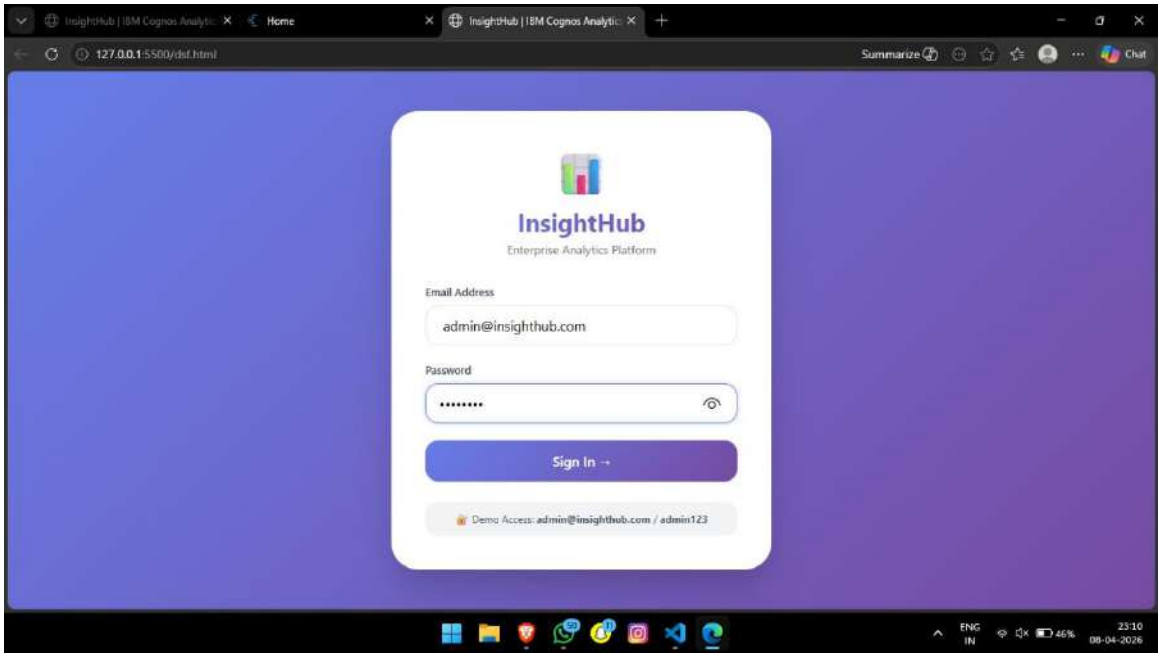
```
        if (validateCredentials(username, password)) {
            sessionStorage.setItem('insightAuth', 'true');
            sessionStorage.setItem('insightUser', username.trim());
            showDashboard(username.trim());
        } else {
            loginError.textContent = '✕ Invalid credentials. Use: admin@insighthub.com / admin123';
            loginError.style.display = 'block';
        }
    });

    // Logout handler
    if (logoutBtn) {
        logoutBtn.addEventListener('click', () => {
            sessionStorage.removeItem('insightAuth');
            sessionStorage.removeItem('insightUser');
            showLogin();
        });
    }

    // Button handlers
    if (openCognosBtn) openCognosBtn.addEventListener('click', openCognosDashboard);
    if (copyUrlBtn) copyUrlBtn.addEventListener('click', copyDashboardUrl);
    if (refreshDatasetBtn) refreshDatasetBtn.addEventListener('click', refreshDatasetInfo);

    // Initialize
    checkAuth();
</script>
</body>
</html>
```

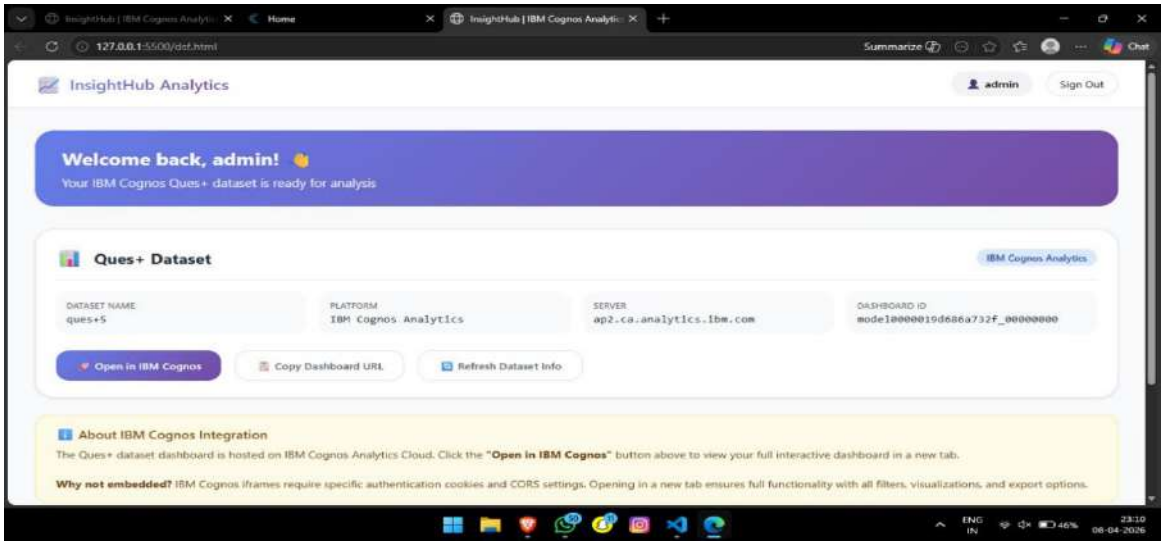
Webpage:

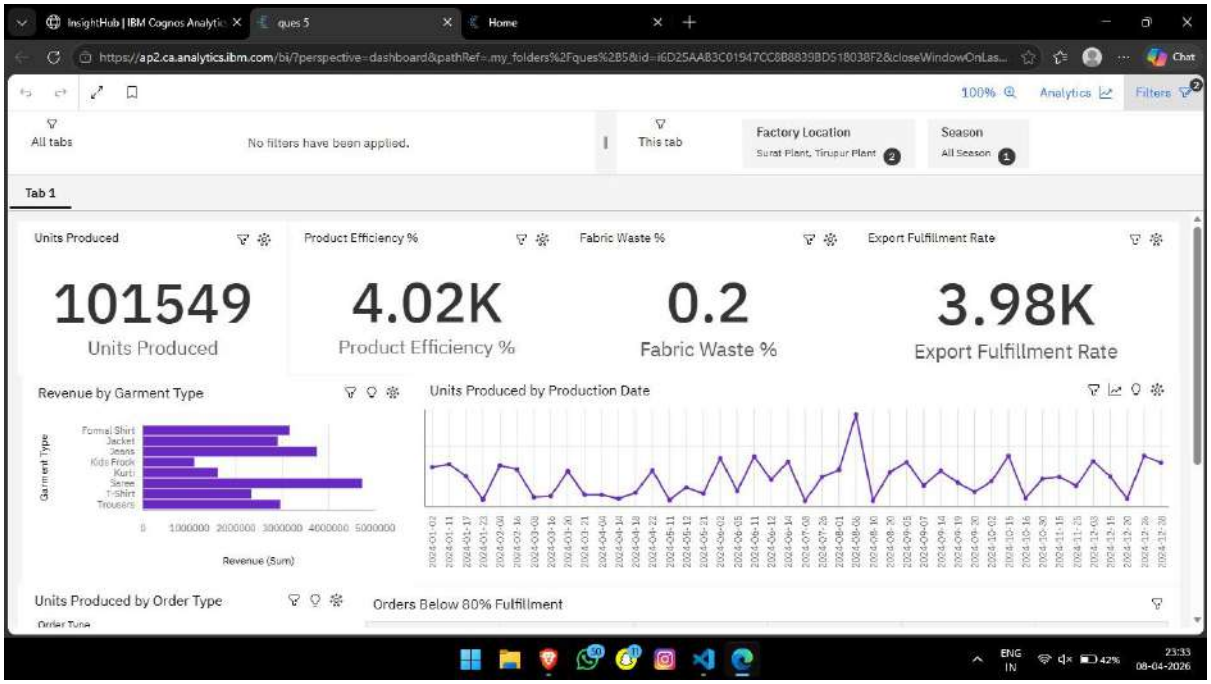
This website has 2 pages:

1. Login Page – where users enter credentials to access the dashboard.
2. Dashboard Page – the main interface showing KPIs, charts, the IBM Cognos iframe, and the shipments data table.

Both pages are built within a single HTML file, with JavaScript toggling visibility between them after successful login.






InsightHub | IBM Cognos Analytic
ques 5
Home
https://ap2.ca.analytics.ibm.com/bi/?perspective=dashboard&pathRef=.my_folders%2Fques%2B5&id=i6D25AAB3C01947CC8B8839BD518038F2&closeWindowOnLas...
Chat
100%
Analytics
Filters
All tabs
No filters have been applied.
This tab
Factory Location
Surat Plant, Tirupur Plant
Season
All Season
Tab 1
Units Produced
101549
Units Produced
Product Efficiency %
4.02K
Product Efficiency %
Fabric Waste %
0.2
Fabric Waste %
Export Fulfillment Rate
3.98K
Export Fulfillment Rate
Revenue by Garment Type
Garment Type
Formal Shirt
Jacket
Jeans
Kids Frock
Kurti
Saree
T-Shirt
Trousers
0
1000000
2000000
3000000
4000000
5000000
Revenue (Sum)
Units Produced by Production Date
Units Produced by Order Type
Order Type
Orders Below 80% Fulfillment
ENG
IN
42%
23:33
08-04-2026

Cognos Story — Quarterly Production & Sales Review

Question:6

Create a Cognos Story for ThreadCraft's Quarterly Production & Sales Review: Scene 1 — Company Overview & Production Summary, Scene 2 — Factory-wise Efficiency Analysis, Scene 3 — Fabric Waste & Rejection Root Causes, Scene 4 — Export Order Performance & Buyer Analysis, Scene 5 — Retail Sales by Season & Category, Scene 6 — Operational Improvement Plan

Procedure

Step 1: Understanding the Requirement

The goal was to create a story with six scenes to present insights clearly. The story includes:

- Company Overview & Production Summary
- Factory-wise Efficiency Analysis
- Fabric Waste & Rejection Root Causes
- Export Order Performance & Buyer Analysis
- Retail Sales by Season & Category
- Operational Improvement Plan

Step 2: Preparing the Data

I checked the dataset and ensured it contains:

- Factory Location
- Garment Type
- Units Planned
- Units Produced
- Units Rejected
- Fabric Type
- Buyer Name
- Order Quantity
- Delivered Quantity
- Retail Category
- Season
- Production Date
- Order Type (Export, Retail, Domestic)

These fields are required to build all scenes.

Step 3: Creating the Story in IBM Cognos Analytics

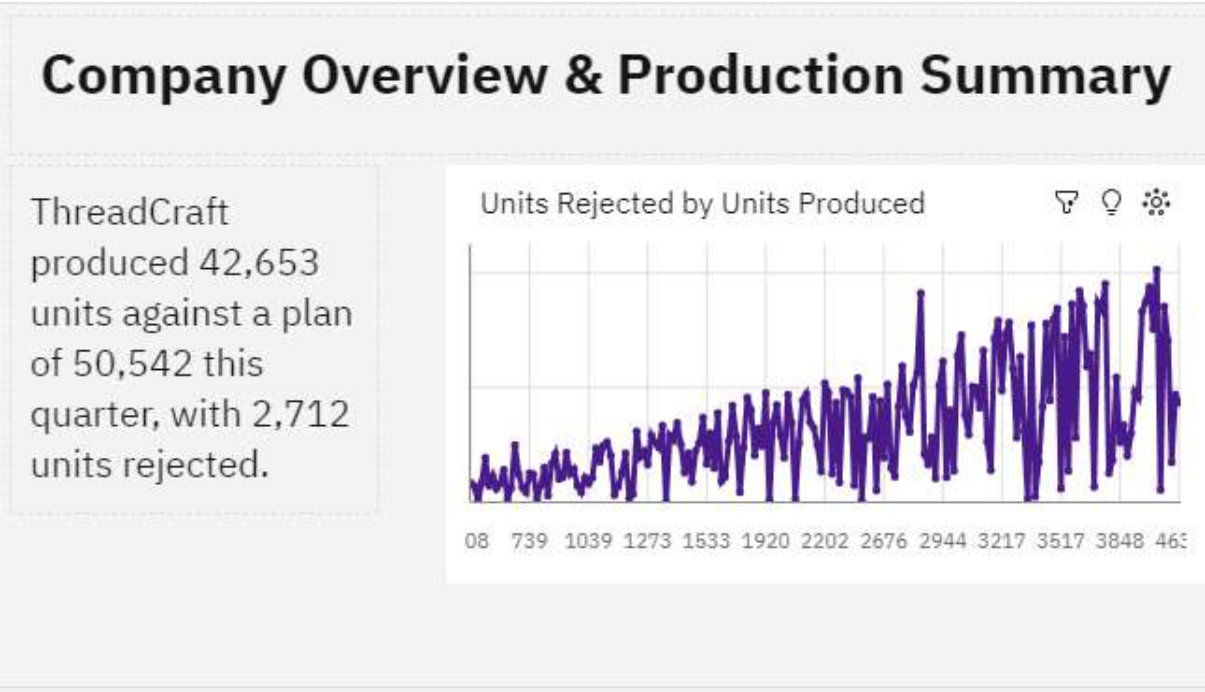
- Opened Cognos Analytics
- Clicked **New → Story**
- Selected **Blank Story**
- Added data module (ThreadCraft dataset)

Step 4: Scene 1 — Company Overview & Production Summary

Added:

- Line chart → Units Produced vs Units Rejected

Insight: This highlights a shortfall in production compared to planned targets and points to efficiency gaps.



Step 5: Scene 2 — Factory-wise Efficiency Analysis

Added:

- Bar chart → Units Produced vs Units Rejected by Factory Location
- Calculated Efficiency % = (Produced / Planned) × 100

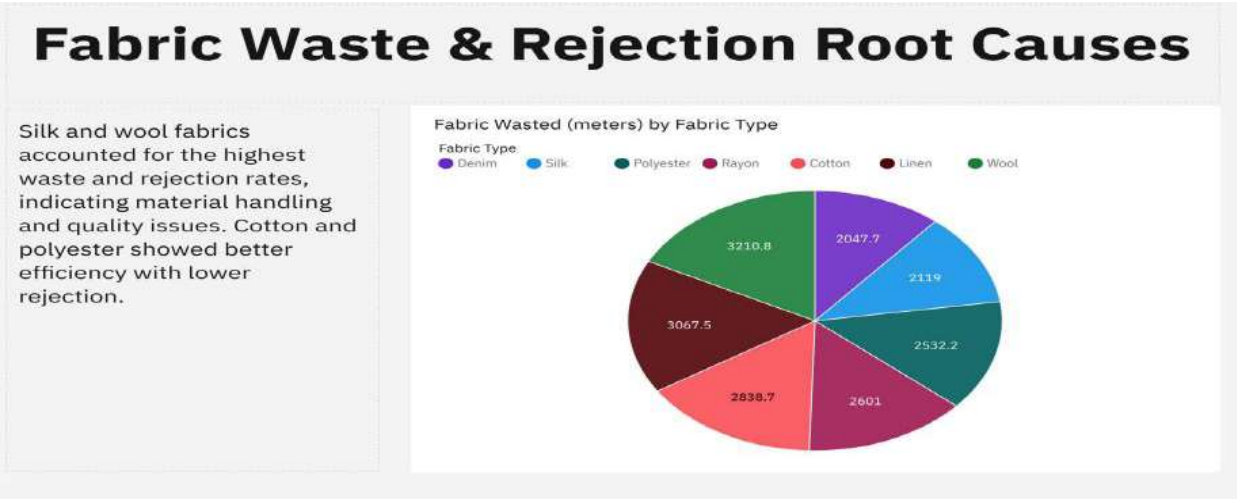
Insight: Tirupur shows higher efficiency compared to Surat.

Step 6: Scene 3 — Fabric Waste & Rejection Root Causes

Added:

- Pie chart → Fabric Waste by Type (Cotton, Silk, Polyester)

Insight: Silk contributes the highest waste, and certain garment types show higher rejection rates.



Step 7: Scene 4 — Export Order Performance & Buyer Analysis

Added:

- **Bar Chart** → *Order Quantity by Buyer Name* (e.g., H&M Sweden, M&S UK, other buyers)

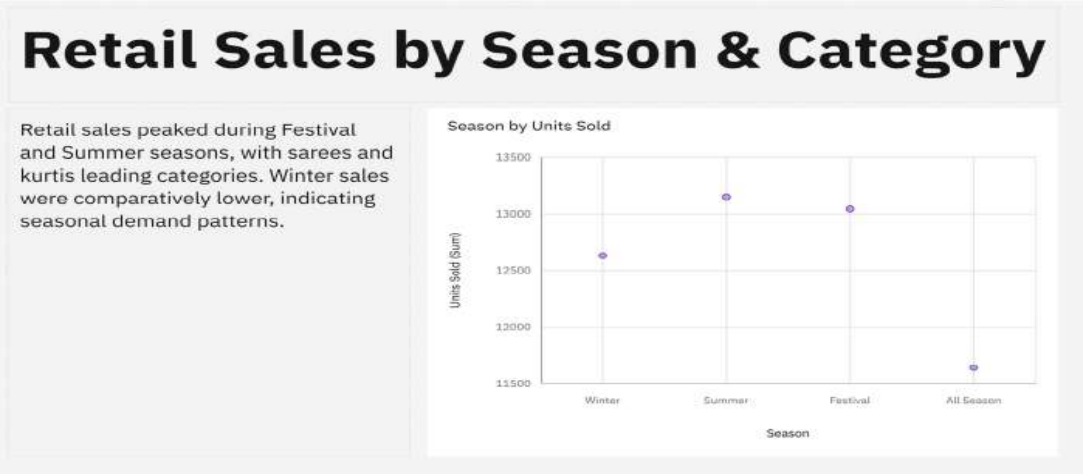
Insight: H&M Sweden and M&S UK were the top export buyers this quarter. While delivery volumes were strong, wool-based orders showed higher rejection rates, impacting overall efficiency.

Step 8: Scene 5 — Retail Sales by Season & Category

Added:

- **Scatter Plot** → X-axis = *Season*, Y-axis = *Units Sold*

Insight: Retail sales peak during festive seasons, with steady demand in summer.

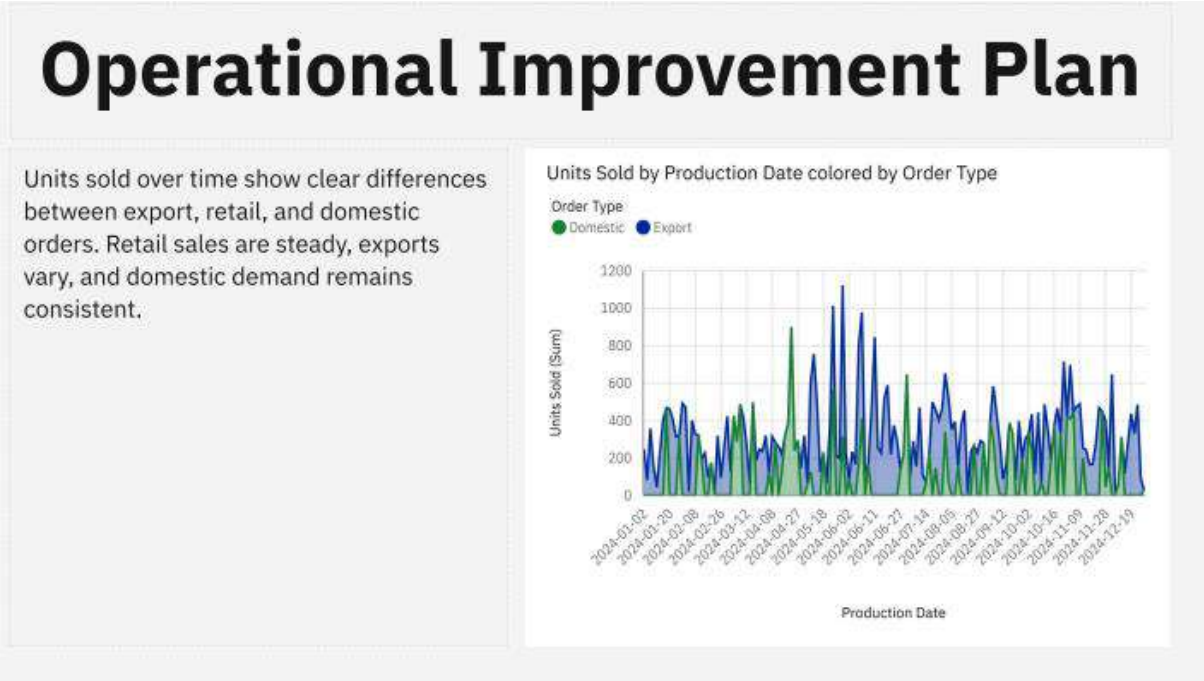


Step 9: Scene 6 — Operational Improvement Plan

Added:

- Area chart → X-axis = Production Date, Y-axis = Units Sold, Color = Order Type
- Filter → Order Type (Export, Retail, Domestic)

Insight: Units sold over time show differences between export, retail, and domestic orders, guiding future improvements.



Step 10: Adding Notes (Annotations)

For each scene, added text notes explaining:

- Key observations
- Problem areas
- Recommendations

Step 11: Final Verification

- Checked all visuals
- Verified calculations
- Ensured filters work properly
- Confirmed story flow is clear

GitHub Repository Link

https://github.com/Prakash-10122005/ThreadCraft_Textiles

Deployment Link

https://prakash-10122005.github.io/ThreadCraft_Textiles/

Conclusion

This project provided a clear view of ThreadCraft's operations using IBM Cognos Analytics. From production efficiency and fabric waste analysis to sales trends and buyer performance, each task built toward a comprehensive understanding of the company's strengths and challenges. The dashboards and story format made insights easy to interpret, while the findings highlighted areas for improvement and growth. Overall, the project shows how data-driven analysis can guide better decisions and support sustainable success.